sudo apt install idle3

Install Node-RED Dashboard:

cd ~/.node-red

npm install node-red-dashboard

Restart Node-RED:

node-red-stop

node-red-start

6. Execution

Now open Node Red in browser:

http://<RaspberryPi-IP>:1880

PRACTICAL-18

1.Aim

Write a program to read the CPU%, Memory% and CPU temperature of Raspberry Pi,

2. Prior Concepts

i) Linux operating system fundamentals

ii) Processor and memory concepts.

3. New Concepts

i) System performance monitoring.

ii) CPU utilization measurement.

iii) Memory utilization analysis.

iv) Using the psutil library.

v) Health monitoring of lot devices.

4. Software Required

i) psutil Library

5. Concept:

CPU Utilization: CPU utilization indicates the percentage of processor resources currently in use.

0% →Idle Processor

100% →Fully Utilized Processor

Vemory Utilization: Memory utilization represents the percentage of RAM currently stupied by running programs and processes.

DU Temperature: CPU temperature helps determine the thermal condition of the processor. Excessive temperature may affect performance and reliability.

Typical Raspberry Pi 4 temperatures:

Temperature	Status
35°C-55°C	Normal
55°C-70°C	Moderate Load
70°C-80°C	High Load
>80°C	Thermal Throttling Possible

6. Procedure

i) Install Python3 on Raspi.

Write the below code in command prompt:

Sudo apt install idle3

Install Libraries

sudo apt update

pip3 install psutil

7. Main Program

```
import psutil
import os
import time
while True:
cpu_percent = psutil.cpu_percent(interval=1)
memory_percent = psutil.virtual_memory().percent
temp = os.popen(
"vcgencmd measure_temp"
).readline()
cpu_temp = temp.replace(
"temp=",
""
).replace(
"C\n",
""
```

```
)
print("\n----------------------")
print(" Raspberry Pi 4 Monitoring")
print("----------------------")
print(
f'CPU Usage: {cpu_percent)%"
)
print(
f'Memory Usage: {memory_percent)%"
)
print(
f'CPU Temperature: {cpu_temp} °C"
)
time.sleep(2)
```

8. Result:

Input IMAGE

```
Raspberry Pi 4 Monitoring
-------------------------
CPU Usage: 18.5%
Memory Usage: 42.3%
```

CPU Temperature: 49.8 °C

Raspberry Pi 4 Monitoring

CPU Usage: 24.1%

Memory Usage: 43.7%

CPU Temperature: 50.1 °C

PRACTICAL-19

1. Aim

Design and implement an embedded access-control prototype that grants entry only when either a whitelisted RFID tag is detected or a legitimate fingerprint is verified, provides real-time visual feedback on an LCD, indicates status via an LED, and pulses a relay active-LOW) to unlock for a fixed interval. The system must log all events (time, method, esult) for auditability.

2. Prior Concepts

i) Concept of Python Programming

ii) Basics of RFID technology

iii) Basics of biometric authentication

vi) SPI communication protocol

v) UART communication protocol

vi) 12C communication protocol